Xu Yuanjian(徐元健)

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Education

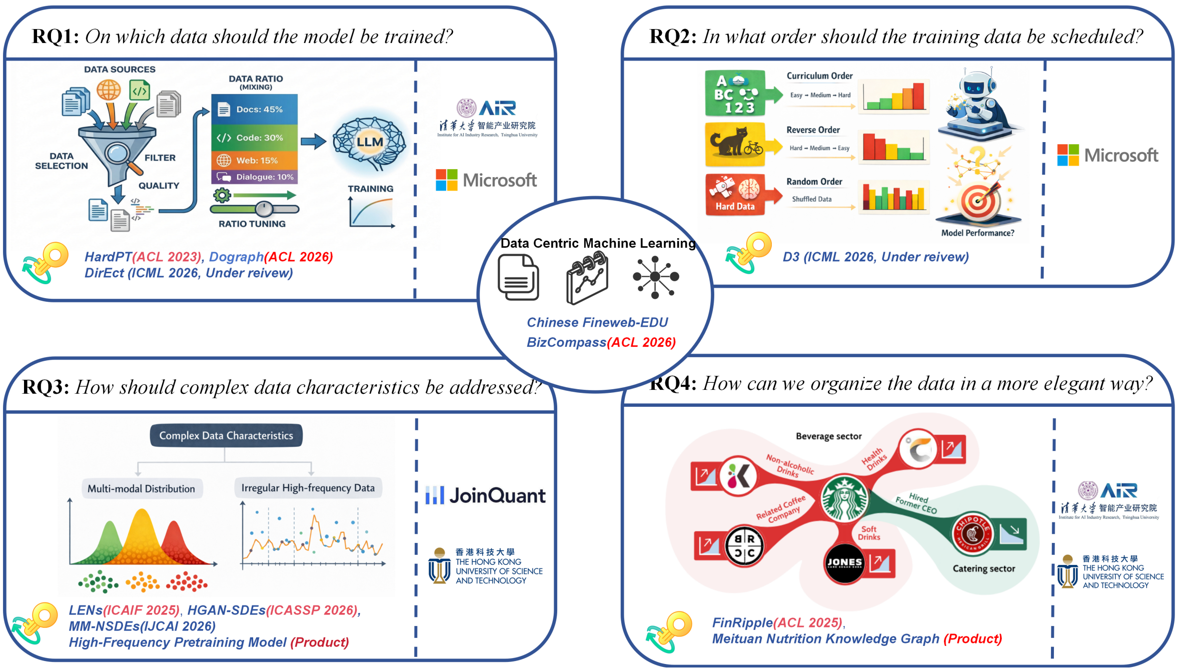
Hong Kong University of Science and Technology, Ph.D. in Fintech (Candidate) 2023.09–Present
Peking University, Master in Computer Science 2020.09–2023.06
Nankai University, Bachelor in Computer Science 2014.09–2018.06

A Brief Introduction

I am a Ph.D. candidate in Computer Science at the Hong Kong University of Science and Technology, Guangzhou campus. I am supervised by Prof. Guang Zhang from HKUST-GZ.
My research interest revolves around **Data-centric Machine Learning**, with a primary focus on **LLMs**. Specifically, my work has systematically investigated the following four dimensions:

- **RQ1 (Data Selection):** Identifying optimal data selection strategies for LLM pre-training.
- **RQ2 (Data Curriculum):** Designing effective data scheduling to maximize model performance.
- **RQ3 (Data Representation):** How to model *complex data characteristics* (e.g. multi-modal data)?
- **RQ4 (Data Orchestration):** How to elegantly organize *heterogeneous data*?

To address these questions, I have published several papers at leading venues including **ACL 2023**, **ACL 2025**, **ACL 2026** (×2), **ACM ICAIF 2025**, and **ICASSP 2026** (oral). Additionally, I have two manuscripts under review at **ICML 2026** (OA > 4) and one at **IJCAI 2026** (passed first round). I also serve as a reviewer for leading conferences such as **NeurIPS 2024/2025/2026**, **ICLR 2024/2025**, and **ICML 2026**.



Selected Publications (First Author)

- [ICML 2026] **Yuanjian Xu**, et al. "D^3: Dynamic Directional Graph-Constrained Data Scheduling for LLM Training." **[CCF A] [CORE A*]** *(Incoming; All reviews are positive 5/5/4/4).*
- [ICML 2026] **Yuanjian Xu**, et al. "Towards Efficient LLMs Annealing with Principled Sample Selection." **[CCF A] [CORE A*]** *(Incoming; All reviews are positive 5/4/4).*
- [ACL 2026] **Yuanjian Xu**, et al. "DoGraph: Towards Domain-Aware LLM Training with Graph-Guided Weighting." **[CCF A] [CORE A*]**.

- [**ACL 2026**] **Yuanjian Xu**, et al. "BizCompass: Benchmarking the Reasoning Capabilities of LLMs in Business Knowledge and Applications." **[CCF A] [CORE A*]**.
- [**ACL 2025**] **Yuanjian Xu**, et al. "FinRipple: Aligning Large Language Models with Financial Market for Event Ripple Effect Awareness." **[CCF A] [CORE A*]**.
- [**ACL 2023**] **Yuanjian Xu**, et al. "Hard Sample Aware Prompt-Tuning." **[CCF A] [CORE A*]**.
- [**ICASSP 2026**] **Yuanjian Xu**, et al. "Hermite Discriminator for Neural SDEs." **[CCF B] [CORE A]** *(Oral)*.
- [**ICAIF 2025**] **Yuanjian Xu**, et al. "LENS: Large Pre-trained Transformer for Exploring Financial Time Series Regularities." *(Leading conference for AI in finance)*.

Under Review & Ongoing Works (First Author)

- [**IJCAI 2026**] **Yuanjian Xu**, et al. "State-Aware Neural Stochastic Differential Equations for Multi-Modal Dynamics." **[CCF A] [CORE A*]** *(Under review; passed first round)*.
- [**NeurIPS 2026**] **Yuanjian Xu**, et al. "FineWeb-Edu-Ultra: A Large-Scale Educational Dataset for Full-Cycle Foundation Model Training." **[CCF A]** *(Ongoing)*.
- [**NeurIPS 2026**] **Yuanjian Xu**, et al. "Data Correction via Logical Dependency Graph Reconstruction." **[CCF A]** *(Ongoing)*.
- [**NeurIPS 2026**] **Yuanjian Xu**, et al. "FinCorpus: A Large-Scale Corpus for Financial Foundation Models." **[CCF A]** *(Ongoing)*.

Research & Teaching Experience

Hong Kong University of Science and Technology, Teaching Assistant, Advanced Statistics

INTERNSHIP EXPERIENCE

AIR, Tsinghua University — Research Intern | 2022.03–2023.03

Worked under the supervision of Prof. Zaiqing Nie. Contributed to the *Meituan Nutrition Knowledge Graph* construction. Investigated *hard sample problems* in NLP and proposed *HardPT*, published at *ACL 2023*.

Microsoft Research Asia (MSRA) — Research Intern | 2024.10–2025.02

Worked under the supervision of Dr. Zhong Li. Focused on *data selection* and *training order optimization* for *large language models*. Proposed the D^3 *method* and an *annealing training framework*, both currently under submission.

Joinquant (Billion-scale quantitative fund) — Research Intern | 2025.02–2025.05

Worked under the supervision of PM Ruixiao. Developed *tick-level generative and representation models* for *high-frequency trading*. Addressed key challenges including *non-equally spaced data* and market randomness.

Skills

Machine Learning: I am well-versed in modern generative models, LLMs, and machine learning.
Mathematical Foundation: I have a relative solid mathematical foundation. During my Ph.D. studies, I passed the qualifying exam in *optimization*, *advanced probability theory*, *advanced statistics*, and *stochastic analysis*.
Financial Theory: I also have exposure to financial theory, having passed *FRM Level 1* and studied *advanced microeconomics* and *advanced asset pricing*.

Honors & Awards

- Award for Excellent Academic Excellence, Peking University — 2021 Certificate No.: H2021000170320
- Air Star Plan, Tsinghua University, Institute for AI Industry Research (AIR) — 2021
- Full Ph.D. Scholarship, Hong Kong University of Science and Technology — 2023–Present
- Tomorrow Star Plan, Microsoft Research Intern — 2025